We remark that compared to the straightforward perturbation bound in Proposition B.8, this approach does not require reversibility nor an explicit condition number bound.

Proof. For all $x, y \in C_k$, by Proposition B.10 applied to X^ε on S and X^0 on C_k,

$$\frac{\pi_k^\varepsilon(x)}{\pi_k^\varepsilon(y)} = \frac{\pi^\varepsilon(x)}{\pi^\varepsilon(y)} = \frac{\mathbb{P}_y(\bar{\tau}_x^\varepsilon < \bar{\tau}_y^\varepsilon)}{\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon)}, \quad \frac{\mu_k(x)}{\mu_k(y)} = \frac{\mathbb{P}_y(\bar{\tau}_x^0 < \bar{\tau}_y^0)}{\mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0)}.$$

Recall that $\mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0) \geq \frac{\nu}{\log M}$ by Lemma C.1 and moreover $\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon) \geq \frac{\nu}{2\log M}$ by repeating the argument in (12). Therefore,

$$\begin{aligned}
\left|\frac{\pi_k^\varepsilon(x)}{\pi_k^\varepsilon(y)} - \frac{\mu_k(x)}{\mu_k(y)}\right| &= \left|\frac{\mathbb{P}_y(\bar{\tau}_x^\varepsilon < \bar{\tau}_y^\varepsilon)}{\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon)} - \frac{\mathbb{P}_y(\bar{\tau}_x^0 < \bar{\tau}_y^0)}{\mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0)}\right| \\
&\leq \frac{\left|\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon) - \mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0)\right| + \left|\mathbb{P}_y(\bar{\tau}_x^\varepsilon < \bar{\tau}_y^\varepsilon) - \mathbb{P}_y(\bar{\tau}_x^0 < \bar{\tau}_y^0)\right|}{\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon)\mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0)} \\
&\leq \frac{4(\log M)^2}{\nu^2} \cdot \widetilde{O}\left(\frac{1}{(\log M)^3}\right).
\end{aligned}$$

By Assumption 1 we have that $\mu_k(y)/\mu_k(x)$ is bounded for all $x, y \in C_k$ and hence

$$\begin{aligned}
\left|\frac{\pi_k^\varepsilon(x)}{\mu_k(x)} - 1\right| &\leq \sum_{y \in C_k} \left|\frac{\pi_k^\varepsilon(x)}{\mu_k(x)}\mu_k(y) - \pi_k^\varepsilon(y)\right| \\
&= \sum_{y \in C_k} \frac{\mu_k(y)}{\mu_k(x)}\pi_k^\varepsilon(y) \left|\frac{\pi_k^\varepsilon(x)}{\pi_k^\varepsilon(y)} - \frac{\mu_k(x)}{\mu_k(y)}\right| \leq \widetilde{O}\left(\frac{1}{\log M}\right).
\end{aligned}$$

□

C.2 Perturbative Analysis of Metastable Chain

We proceed to study the behavior of the meta-chain $X_\star^\varepsilon$ with transition probabilities $q_\star^\varepsilon$ defined in (2). It can be shown that $X_\star^\varepsilon$ is asymptotically reversible with respect to the measure induced by π^ε:

Proposition C.8. *For all $k, \ell \in [K]$ with $k \neq \ell$ it holds that*

$$\frac{\pi^\varepsilon(C_k)q_\star^\varepsilon(C_\ell|C_k)}{\pi^\varepsilon(C_\ell)q_\star^\varepsilon(C_k|C_\ell)} = 1 + \widetilde{O}\left(\frac{1}{\log M}\right).$$

Proof. First note that for $x \in C_k$, $y \in C_\ell$ with $k \neq \ell$, by Proposition C.5,

$$0 \leq 1 - \frac{\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon)}{\mathbb{P}_x(\bar{\tau}_{C_\ell}^\varepsilon < \bar{\tau}_x^\varepsilon)}$$

$$
\begin{aligned}
&= \frac{1}{\mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x)} \sum_{z \in C_\ell} \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x, X^\varepsilon_{\bar{\tau}^\varepsilon_{C_\ell}} = z) \mathbb{P}_z(\tau^\varepsilon_x < \tau^\varepsilon_y) \\
&\leq \frac{1}{\mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x)} \sum_{z \in C_\ell} \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x, X^\varepsilon_{\bar{\tau}^\varepsilon_{C_\ell}} = z) \cdot \sup_{z \in C_\ell} \left| \mathbb{P}_z(\tau^\varepsilon_x < \tau^\varepsilon_y) - \mathbb{P}_z(\tau^0_x < \tau^0_y) \right| \\
&\leq \widetilde{O}\left(\frac{1}{(\log M)^3} \right).
\end{aligned}
$$

By the definition of $q^\varepsilon_\star$, the coupling equation (14) and Corollary C.7, it follows that

$$
\begin{aligned}
\pi^\varepsilon(C_k) q^\varepsilon_\star(C_\ell | C_k) &= \pi^\varepsilon(C_k) \sum_{x \in C_k} \mu_k(x)^2 \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \\
&= \pi^\varepsilon(C_k) \sum_{x \in C_k} \pi^\varepsilon_k(x) \mu_k(x) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \\
&\qquad + \pi^\varepsilon(C_k) \sum_{x \in C_k} \mu_k(x)^2 \left(1 - \frac{\pi^\varepsilon_k(x)}{\mu_k(x)} \right) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \\
&= \sum_{x \in C_k} \pi^\varepsilon(x) \mu_k(x) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) + \pi^\varepsilon(C_k) q^\varepsilon_\star(C_\ell | C_k) \cdot \widetilde{O}\left(\frac{1}{\log M} \right).
\end{aligned}
$$

We expand the first term further as

$$
\begin{aligned}
&\sum_{x \in C_k} \pi^\varepsilon(x) \mu_k(x) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \\
&= \sum_{x \in C_k} \sum_{y \in C_\ell} \pi^\varepsilon(x) \mu_k(x) \mu_\ell(y) \mathbb{P}_x(\bar{\tau}^\varepsilon_y < \bar{\tau}^\varepsilon_x) \\
&\qquad + \sum_{x \in C_k} \sum_{y \in C_\ell} \pi^\varepsilon(x) \mu_k(x) \mu_\ell(y) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \left(1 - \frac{\mathbb{P}_x(\bar{\tau}^\varepsilon_y < \bar{\tau}^\varepsilon_x)}{\mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x)} \right) \\
&= \sum_{x \in C_k} \sum_{y \in C_\ell} \pi^\varepsilon(x) \mu_k(x) \mu_\ell(y) \mathbb{P}_x(\bar{\tau}^\varepsilon_y < \bar{\tau}^\varepsilon_x) \\
&\qquad + \sum_{x \in C_k} \pi^\varepsilon(x) \mu_k(x) \mathbb{P}_x(\bar{\tau}^\varepsilon_{C_\ell} < \bar{\tau}^\varepsilon_x) \cdot \widetilde{O}\left(\frac{1}{(\log M)^3} \right).
\end{aligned}
$$

Together, we have shown that

$$
\pi^\varepsilon(C_k) q^\varepsilon_\star(C_\ell | C_k) = \sum_{x \in C_k} \sum_{y \in C_\ell} \pi^\varepsilon(x) \mu_k(x) \mu_\ell(y) \mathbb{P}_x(\bar{\tau}^\varepsilon_y < \bar{\tau}^\varepsilon_x) \cdot \left(1 + \widetilde{O}\left(\frac{1}{\log M} \right) \right),
$$

and by symmetry

$$
\pi^\varepsilon(C_\ell) q^\varepsilon_\star(C_k | C_\ell) = \sum_{x \in C_k} \sum_{y \in C_\ell} \pi^\varepsilon(y) \mu_k(x) \mu_\ell(y) \mathbb{P}_y(\bar{\tau}^\varepsilon_x < \bar{\tau}^\varepsilon_y) \cdot \left(1 + \widetilde{O}\left(\frac{1}{\log M} \right) \right).
$$

Finally, since

$$\sum_{x\in C_k}\sum_{y\in C_\ell}\mu_k(x)\mu_\ell(y)\cdot\pi^\varepsilon(x)\mathbb{P}_x(\bar\tau_y^\varepsilon<\bar\tau_x^\varepsilon)=\sum_{x\in C_k}\sum_{y\in C_\ell}\mu_k(x)\mu_\ell(y)\cdot\pi^\varepsilon(y)\mathbb{P}_y(\bar\tau_x^\varepsilon<\bar\tau_y^\varepsilon)$$

due to Proposition B.10, we conclude the desired statement. □

Together with Assumptions 1, 4 and (14), this immediately implies:

Corollary C.9. *For all $k\in[K]$ and $x\in S$ it holds that $\pi^\varepsilon(C_k)=\Theta(1/K)$ and $\pi^\varepsilon(x)=\Theta(1/KM)$.*

Moreover, $q_\star^\varepsilon(\cdot|C_k)$ serves as an approximation of the escape probabilities from any $x\in C_k$, weighted by the stationary measure.

Proposition C.10. *For $k,\ell\in[K]$ with $k\neq\ell$ it holds that*

$$\sup_{x\in C_k}\left|\frac{\mu_k(x)\mathbb{P}_x(\bar\tau_{C_\ell}^\varepsilon<\bar\tau_x^\varepsilon)}{q_\star^\varepsilon(C_\ell|C_k)}-1\right|=\widetilde{O}\left(\frac{1}{\log M}\right)\tag{15}$$

and

$$\sup_{x\in C_k,y\in C_\ell}\left|\frac{\mu_k(x)\mathbb{P}_x(\bar\tau_y^\varepsilon<\bar\tau_x^\varepsilon)}{q_\star^\varepsilon(C_\ell|C_k)}-1\right|=\widetilde{O}\left(\frac{1}{\log M}\right).\tag{16}$$

Proof. Similarly to the proof of Proposition C.8, for any $y\in C_\ell$ we can successively transform

$$\begin{aligned}\mu_k(x)\mathbb{P}_x(\bar\tau_{C_\ell}^\varepsilon<\bar\tau_x^\varepsilon)&=\mu_k(x)\mathbb{P}_x(\bar\tau_y^\varepsilon<\bar\tau_x^\varepsilon)\cdot\left(1+\widetilde{O}\left(\frac{1}{\log M}\right)\right)\\&=\pi_k^\varepsilon(x)\mathbb{P}_x(\bar\tau_y^\varepsilon<\bar\tau_x^\varepsilon)\cdot\left(1+\widetilde{O}\left(\frac{1}{\log M}\right)\right)\\&=\pi_k^\varepsilon(y)\mathbb{P}_y(\bar\tau_x^\varepsilon<\bar\tau_y^\varepsilon)\cdot\left(1+\widetilde{O}\left(\frac{1}{\log M}\right)\right)\\&=\pi_k^\varepsilon(y)\mathbb{P}_y(\bar\tau_{C_k}^\varepsilon<\bar\tau_y^\varepsilon)\cdot\left(1+\widetilde{O}\left(\frac{1}{\log M}\right)\right).\end{aligned}$$

Since the last term is independent of x, we also have

$$q_\star^\varepsilon(C_\ell|C_k)=\sum_{x\in C_k}\mu_k(x)^2\mathbb{P}_x(\bar\tau_{C_\ell}^\varepsilon<\bar\tau_x^\varepsilon)=\pi_k^\varepsilon(y)\mathbb{P}_y(\bar\tau_{C_k}^\varepsilon<\bar\tau_y^\varepsilon)\cdot\left(1+\widetilde{O}\left(\frac{1}{\log M}\right)\right),$$

verifying (15). The proof for (16) is identical. □

As a corollary, we obtain the promised justification of Assumption 4.

Corollary C.11. *If there exists a sparse edge from C_k to C_ℓ, it holds that $q_\star^\varepsilon(C_\ell|C_k)=\widetilde{\Omega}(\varepsilon/M)$.*